

References: Read Reactor Theory Manual Chapter 1: §1.4
On Moodle:

- *Lecture Module #4 slides and K-Effective, Reactivity & Criticality Handout PDF*
- *Lab #3 – Normal Reactor Startup PDF*

1. Give 3 examples of nuclear reactions yielding “Source Neutrons” in the reactor core:
2. List 2 “SOURCES” and 4 “SINKS” of **fission** reaction induced neutrons (**not** source neutrons) in the reactor core, AND give the factor in the 6-FACTOR FORMULA that accounts for each:

SOURCES:		“6-Factor”
A.		
B.		
SINKS:		
A.		
B.		
C.		
D.		

3. 6 Factor Formula Definitions - *MATCHING*

<u>Factor:</u>	<u>Definitions:</u>
_____ η	A. Ratio of number of fast neutrons that start to slow down (moderate), to the number of fast neutrons from fast and thermal fission.
_____ ε	B. Ratio of the number of thermal neutrons that are absorbed in the core, to the number of neutrons that reach thermal energy.
_____ p	C. Ratio of fast neutrons from both fast & thermal fission, to the number of fast neutrons from thermal fission alone.
_____ f	D. Ratio of the number of neutrons that reach thermal energy, to the number of neutrons that started to slow down (moderate).
_____ $\mathcal{L}_{\text{TH}}$	E. Ratio of the number of thermal neutrons that are absorbed by the fuel, to the number of thermal neutrons that are absorbed in the core.
_____ $\mathcal{L}_{\text{F}}$	F. Ratio of the number of neutrons generated from thermal fission, to the number of thermal neutrons absorbed in the fuel.

4. Fill in the blanks with either a value or a range of values where appropriate (i.e. $0 < \rho/\beta \leq 0.5$ etc...) using the correct units. (Hint: if $\rho = \$1 = 730 \text{ pcm}$ then $K_{eff} = 1.0073$)

If the reactor is:	K_{eff} is:	Reactivity ρ/β (\$) is:
Super-prompt critical:		
Prompt-critical:		
Super delayed critical:		
Critical:		
Sub-critical:		

Lab Questions

REF: Reactor Startup Lab Handout

1. What are the three characteristics of subcritical multiplication that were observed while taking the reactor from shutdown to (super) critical?
2. What are two indications on the reactor console that the reactor is super-critical?
3. Which indicates a faster rate of power increase, a 10 second or a 30 second Period τ (sec)?
4. According to INPO, what is the first thing you should do before performing a reactor operations procedure?